

The combustion of fossil fuels produces flue gases containing carbon dioxide. One approach for reducing these emissions is post-combustion CO₂ capture, in which the flue gas is contacted with an amine-based liquid absorbent inside a packed column. The packing increases the contact area between the gas and liquid phases and promotes the transfer of CO₂ to the absorbent. The absorption process is shown in Figure 1.

An energy company is evaluating the design of a new packed absorption column. For this preliminary study, a synthetic dataset has been generated that relates the operating conditions and column dimensions to the gas-side volumetric mass-transfer coefficient (kG_a) and the CO₂ capture efficiency (η).

The dataset contains the following variables:

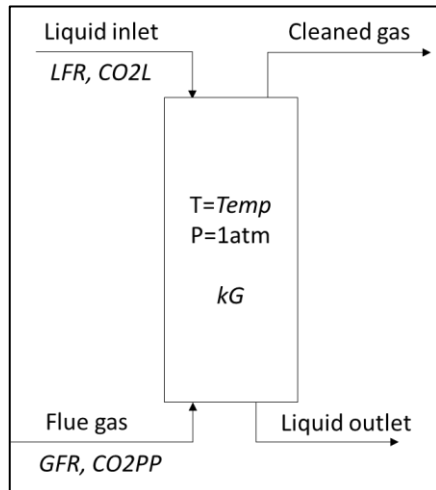


Figure 1. Absorption process schematic representation

- The gas flow rate per unit column area (GFR in the dataset), in $m^3/(m^2 \cdot h)$.
- The liquid flow rate per unit column area (LFR in the dataset), in $L/(m^2 \cdot h)$.
- The operating temperature ($Temp$ in the dataset), in $^{\circ}C$.
- The initial CO₂ loading of the liquid at the column inlet ($CO2alpha$ in the dataset), in $mol\ CO_2/mol\ amine$.
- The CO₂ partial pressure in the gas at the column inlet ($CO2pp$ in the dataset), in kPa .
- The height of the packed section ($Height$ in the dataset), in m .
- The gas-side volumetric mass-transfer coefficient (kG_a in the dataset), in $kmol \cdot m^{-2} \cdot h^{-1} \cdot kPa^{-1}$.
- The CO₂ capture efficiency (η in the dataset), in $mol\ CO_2\ captured/mol\ CO_2\ entering$.

Assume that the gas and liquid flow rates remain constant along the column.

Construct and evaluate a machine-learning surrogate of the absorption column by completing the following tasks:

- Load the dataset and inspect its dimensions, data types, summary statistics, and missing values. Visualize the distribution of each variable using histograms.
- Select GFR , LFR , $Temp$, $CO2alpha$, $CO2pp$, and $Height$ as the model inputs. Select kG_a and η as the model outputs, in that order, and convert both selections into NumPy arrays.
- Randomly reserve 20% of the observations as an unseen test set. Use a random seed of 42. Standardize the inputs and outputs using separate standard scalers fitted only on the training data.
- Set the Keras random seed to 42 and construct a sequential artificial neural network with six inputs, three hidden layers containing 10 neurons each, ReLU activation in the hidden layers, and two output neurons with linear activation.
- Compile the model using the Adam optimizer and mean squared error as the loss function. Display the model summary and verify its architecture.
- Train the model for 500 epochs using a batch size of 32. Reserve 20% of the training observations for validation and monitor the training and validation losses.
- Predict kG_a and η for the test observations and convert the predictions back to their original physical scale.
- Evaluate the predictive performance by calculating R^2 and RMSE separately for kG_a and η . Construct a parity plot for each output and include the line representing ideal predictions.

After evaluating the model, use it to investigate the design of a column that can process the flue-gas stream described in Table 1.

Table 1. Flue-gas physical characteristics

Total gas flow rate	20,000 m ³ /h
Inlet CO ₂ partial pressure	10 kPa
Initial CO ₂ loading of the liquid	0.1 mol CO ₂ /mol amine

Use differential evolution to determine the operating conditions and packed height that maximize the neural network's predicted kGa. Complete the following steps:

- Optimize GFR, LFR, Temp, and Height while keeping CO₂alpha equal to 0.1 mol CO₂/mol amine and CO₂pp equal to 10 kPa.
- Use the minimum and maximum values in the training data as the bounds for each decision variable. Set the differential-evolution random seed to 42 and retain the final polishing step.
- Report the selected gas flow rate per unit area, liquid flow rate per unit area, operating temperature, packed height, predicted kGa, and predicted eta.
- Use an if-else statement to verify that the predicted capture efficiency lies between 0 and 1.
- Calculate the required column cross-sectional area and diameter for a total gas flow rate of 20,000 m³/h. Calculate the corresponding total liquid flow rate from the selected LFR.

Treat the optimized result as a model-generated candidate rather than a validated process design. Check whether the proposed condition is close to the limits of the available data and identify what additional process knowledge or validation would be required before practical use.

Optional extension: train a multi-output linear-regression model using the same training and test observations. Calculate R² and RMSE separately for kGa and eta and compare its performance with that of the neural network.